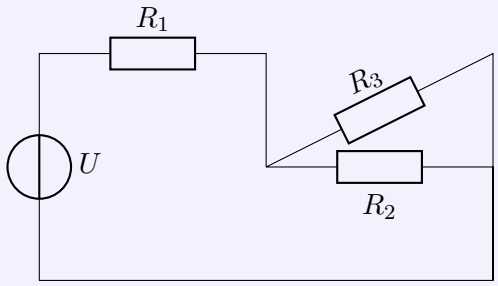
Exercise Sheet #3

DC 2

Exercise 1: Network Analysis 1

Given the circuit with $R_1 = 6\,\Omega$, $R_2 = 12\,\Omega$, $R_3 = 4\,\Omega$ and a voltage source of $24\,\mathrm{V}$:



Determine:

1. the total source current,
2. the branch currents through R_2 and R_3,
3. the power dissipated in R_3.

Solution

$$R_{23} = \frac{12 \cdot 4}{16} = 3\,\Omega$$

$$R = 6 + 3 = 9\,\Omega$$

$$I = \frac{24}{9} = 2.67\,\mathrm{A}$$

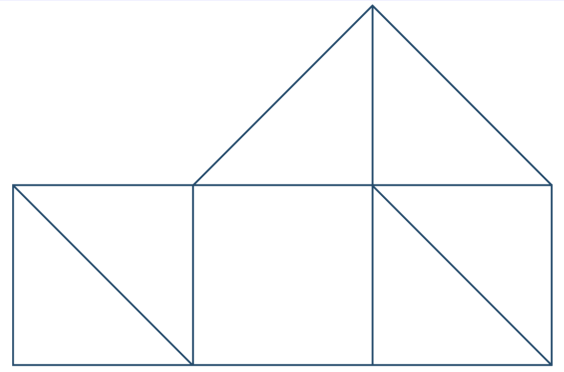
$$U_{23} = I \cdot 3 = 8\,\mathrm{V}$$

$$I_2 = \frac{8}{12} = 0.67\,\mathrm{A}, \quad I_3 = \frac{8}{4} = 2\,\mathrm{A}$$

$$P_3 = I_3^2 R_3 = 16\,\mathrm{W}$$

Exercise 2: Topology

Find the number of nodes, branches, and loops.

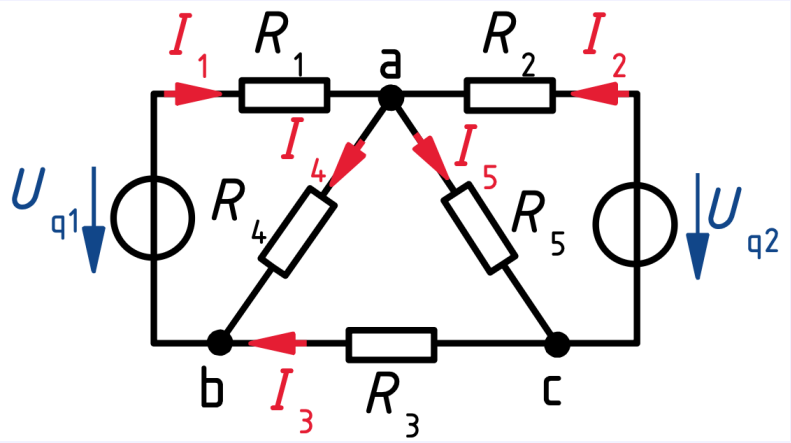


Solution

There are $n = 8$ nodes, corresponding to the junctions.
There are $b = 14$ branches, corresponding to the individual segments connecting two nodes.
There are overall $l = 15$ geometric loops, i.e., closed paths. But only $m = b - n + 1 = 7$ of these closed paths are independent loops. One could choose the 7 meshes (smallest possible loops, not enclosing any other loop) for an electrical network analysis, but other combinations are also possible.

Exercise 3: Network Analysis 2

The following network contains the resistors $R_1 = 1\,\Omega$, $R_2 = 2\,\Omega$, $R_3 = 5\,\Omega$, $R_4 = 25\,\Omega$, and $R_5 = 40\,\Omega$, as well as the source voltages $U_{q1} = 16.2\,\mathrm{V}$ and $U_{q2} = 11.4\,\mathrm{V}$. All branch currents are to be determined.



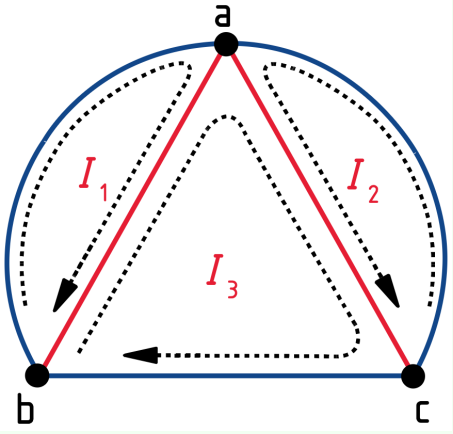
Solution

This network has $n = 3$ nodes, so there are $n-1 = 2$ linearly independent node equations. Picking nodes B and C:

$$I_3 + I_4 - I_1 = 0$$

$$I_5 - I_2 - I_3 = 0$$

The number of branches is 5, so there are $5 - (3 - 1) = 3$ linearly independent mesh equations. Graph of the network with a selected complete tree (marked in red):



The branch currents in the connecting branches, i.e., I_1, I_2, and I_3, are chosen as mesh currents. The mesh equations are set up by traversing the meshes in the direction of the counting arrows of the mesh currents:

$$\begin{aligned} M_1: &\quad R_1 I_1 + R_4 I_4 - U_{q1} = 0 \\ M_2: &\quad R_2 I_2 + R_5 I_5 - U_{q2} = 0 \\ M_3: &\quad R_3 I_3 - R_4 I_4 + R_5 I_5 = 0 \end{aligned}$$

Resulting system of equations as matrix equation:

$$\begin{bmatrix} -1 & 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 1 \\ R_1 & 0 & 0 & R_4 & 0 \\ 0 & R_2 & 0 & 0 & R_5 \\ 0 & 0 & R_3 & -R_4 & R_5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ U_{q1} \\ U_{q2} \\ 0 \end{bmatrix}.$$

Substituting numerical values gives:

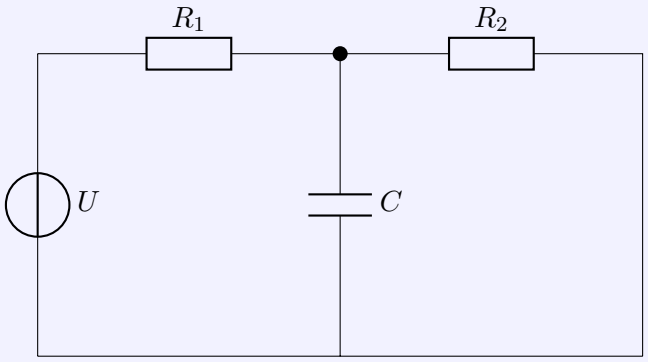
$$\begin{bmatrix} -1 & 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 1 \\ 1 & 0 & 0 & 25 & 0 \\ 0 & 2 & 0 & 0 & 40 \\ 0 & 0 & 5 & -25 & 40 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 16.2 \\ 11.4 \\ 0 \end{bmatrix}.$$

...

$$I_1 = 1.2\,\text{A}, \qquad I_2 = -0.3\,\text{A}, \qquad I_3 = 0.6\,\text{A}, \qquad I_4 = 0.6\,\text{A}, \qquad I_5 = 0.3\,\text{A}.$$

Exercise 4: Nodal Analysis

Given the DC network with $U = 20\,\text{V}$, $R_1 = 5\,\Omega$, $R_2 = 10\,\Omega$, $C = 100\,\mu\text{F}$:



The lower conductor is chosen as reference potential (ground). The steady-state condition ($t \to \infty$) is assumed.
Determine:

1. the node voltage at the marked node,
2. the voltage across the capacitor,
3. the current through the capacitor,
4. the energy stored in the capacitor.

Solution

$$\frac{V-U}{R_1} + \frac{V-0}{R_2} = 0$$

$$2(V-20) + V = 0$$

$$V = \frac{40}{3} \approx 13.33\,\mathrm{V}$$

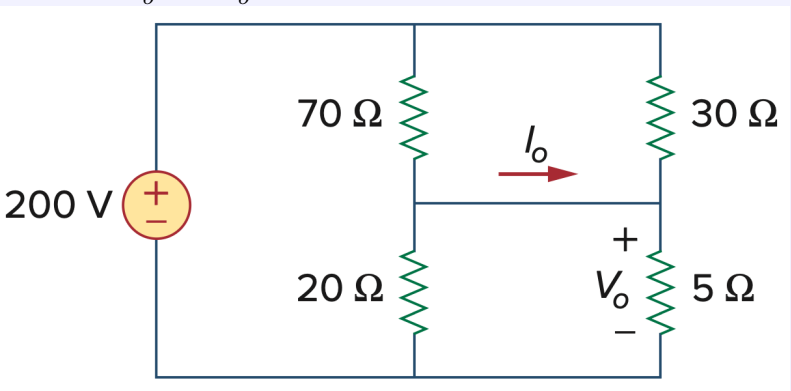
$$U_C = V - 0 = 13.33\,\mathrm{V}$$

$$i_C = C\frac{du_C}{dt} = 0$$

$$W = \frac{1}{2}CU_C^2 \approx 8.9\,\mathrm{mJ}$$

Exercise 5: Node Connection

Calculate I_o and V_o.



Solution

A node is a point where two or more circuit elements connect (electrically the same point). A branch is a path that connects two nodes and contains a circuit element (resistor, source, etc.).
Here, there are three main nodes:

- Top node: connected to the +200 V source, top of the 70 Ω and 30 Ω resistors.
- Bottom node: 0 V reference.
- Middle node: connection between two branches. On the left, the branch formed by the 70 Ω and 20 Ω. And on the right, the branch formed by the 30 Ω and 5 Ω.

The middle horizontal wire (with I_o) is part of the middle node, but because there's a potential difference between the left and right parts of that node before it's connected, current flows through that connection — hence we analyze I_o as if it's a branch current. As the output voltage V_o is the voltage across the 5 Ω resistor, the output voltage equals the middle node voltage.

Currents from the top node to the middle node:

$$I_{70} = \frac{200\,\text{V} - V_o}{70\,\Omega}, \qquad I_{30} = \frac{200\,\text{V} - V_o}{30\,\Omega}$$

Currents from the middle node to the bottom node:

$$I_{20} = \frac{V_o}{20\,\Omega}, \qquad I_5 = \frac{V_o}{5\,\Omega}$$

Applying KCL at the middle node:

$$\frac{200\,\text{V} - V_o}{70\,\Omega} + \frac{200\,\text{V} - V_o}{30\,\Omega} = \frac{V_o}{20\,\Omega} + \frac{V_o}{5\,\Omega}$$

Simplify:

$$\frac{200\,\text{V} - V_o}{21\,\Omega} = \frac{V_o}{4\,\Omega}$$

$$\Rightarrow 800\,\text{V} - 4 \cdot V_o = 21 \cdot V_o$$

$$\Rightarrow \boxed{V_o = 32\ \text{V}}$$

Now we use KCL to compute the current I_o (defined to the right):

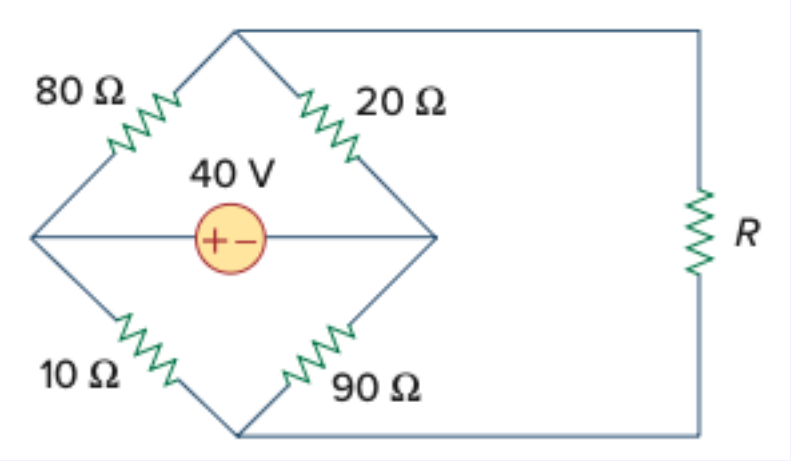
$$I_o = I_{70} - I_{20} = \frac{200\,\text{V} - 32\,\text{V}}{70\,\Omega} - \frac{32\,\text{V}}{20\,\Omega} = \boxed{0.8\,\text{A}}$$

Exercise 6: Thévenin and Power Matching

The variable resistor R is adjusted until it absorbs the maximum power from the circuit.

a) Calculate the value of R for maximum power.

b) Determine the maximum power absorbed by R.



Solution

a) First, we calculate the Thévenin equivalent seen from the terminals of R.

Let node D (left of 40 V source) be at 40 V, and node B (right of source) be at 0 V.

For the upper node A:

$$\frac{V_A - 40}{80} + \frac{V_A - 0}{20} = 0 \Rightarrow 5V_A - 40 = 0 \Rightarrow V_A = 8\,\text{V}$$

For the lower node C:

$$\frac{V_C - 40}{10} + \frac{V_C - 0}{90} = 0 \Rightarrow 10V_C - 360 = 0 \Rightarrow V_C = 36\,\text{V}$$

Thus, the open-circuit (Thévenin) voltage is:

$$V_{\text{th}} = |V_A - V_C| = |8 - 36| = 28\,\text{V}$$

Now we deactivate the 40 V source (replace by a short circuit):

$$R_A = (80\,\Omega \parallel 20\,\Omega) = \frac{80 \cdot 20}{80 + 20} = 16\,\Omega$$

$$R_C = (10\,\Omega \parallel 90\,\Omega) = \frac{10 \cdot 90}{10 + 90} = 9\,\Omega$$

Since R_A and R_C are in series,

$$R_{\text{th}} = R_A + R_C = 16 + 9 = 25\,\Omega.$$

For maximum power: $R = R_{\text{th}} = 25\,\Omega$

b) The circuit can be represented by a Thévenin voltage source V_{th} in series with a Thévenin resistance R_{th} and a load R_L.

$$I = \frac{V_{\text{th}}}{R_{\text{th}} + R_L}$$

Power in the load:

$$P = I^2 R_L = \frac{V_{\text{th}}^2 R_L}{(R_{\text{th}} + R_L)^2}$$

Substitute $R_L = R_{\text{th}}$ into the power equation:

$$P_{\max} = \frac{V_{\text{th}}^2 R_{\text{th}}}{(R_{\text{th}} + R_{\text{th}})^2} = \frac{V_{\text{th}}^2}{4R_{\text{th}}} = \frac{28^2}{4 \cdot 25} = \frac{784}{100} = 7.84\,\text{W}$$

Exercise 7: Efficiency

For a real voltage source with an internal resistance and a connected load, the efficiencies given below apply. For each case, calculate what fraction of the maximum possible power is absorbed by the load.

a) $\eta = 75\%$

b) $\eta = 90\%$

Solution

Efficiency η is defined as the ratio of power delivered to the load to the total power supplied by the source.
For a Thévenin source (V_{th}, R_{th}) with a load R_L, the efficiency is:

$$\eta = \frac{P_L}{P_L + P_{\text{th}}} = \frac{R_L}{R_L + R_{\text{th}}}$$

At maximum power ($R_L = R_{\text{th}}$):

$$P_{L,\max} = \frac{V_{\text{th}}^2}{4R_{\text{th}}}$$

And:

$$P_L = I^2 \cdot R_L = \frac{V_{\text{th}}^2}{(R_{\text{th}} + R_L)^2} \cdot R_L = \frac{V_{\text{th}}^2 R_L}{(R_{\text{th}} + R_L)^2}$$

Hence, the power ratio is:

$$\frac{P_L}{P_{L,\max}} = \frac{\frac{V_{\text{th}}^2 R_L}{(R_{\text{th}} + R_L)^2}}{\frac{V_{\text{th}}^2}{4R_{\text{th}}}} = 4R_{\text{th}} \cdot \frac{R_L}{(R_{\text{th}} + R_L)^2}$$

a) For $\eta = 75\%$:

$$0.75 = \frac{R_L}{R_L + R_{\text{th}}}$$
$$\Rightarrow R_L = 3R_{\text{th}}$$
$$\Rightarrow \frac{P_L}{P_{L,\max}} = 4R_{\text{th}} \cdot \frac{3R_{\text{th}}}{(R_{\text{th}} + 3R_{\text{th}})^2} = \frac{12R_{\text{th}}^2}{(4R_{\text{th}})^2} = 0.75$$

Even when efficiency increases to 75%, the load still receives 75% of the maximum possible power. Thus, higher efficiency is achieved at the cost of only a moderate reduction in power delivered to the load.

b) For $\eta = 90\%$:

$$0.9 = \frac{R_L}{R_L + R_{\text{th}}}$$
$$\Rightarrow R_L = 9R_{\text{th}}$$
$$\Rightarrow \frac{P_L}{P_{L,\max}} = 4R_{\text{th}} \cdot \frac{9R_{\text{th}}}{(R_{\text{th}} + 9R_{\text{th}})^2} = \frac{36R_{\text{th}}^2}{(10R_{\text{th}})^2} = 0.36$$

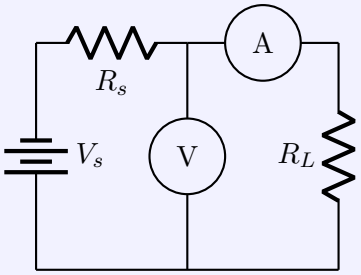
So as efficiency rises, delivered power drops sharply.

Exercise 8: Power Measurement

Calculate the effect of voltmeter and ammeter placement on the measured power in a DC resistive circuit with the parameters and configuratons below. Compare the measured power to the true power dissipated in the load resistor for both configurations and for the two different load resistances.

- Source voltage: $V_s = 12\,\mathrm{V}$
- Source internal resistance: $R_s = 1\,\Omega$
- Voltmeter internal resistance: $R_v = 1000\,\Omega$
- Ammeter internal resistance: $R_a = 0.1\,\Omega$
- Load resistance (two cases):
  - Low load resistance: $R_L = 1\,\Omega$
  - High load resistance: $R_L = 100\,\Omega$

1. Configuration A (Current-corrected):
   Voltmeter connected across the source, ammeter in series with the load.



2. Configuration B (Voltage-corrected):
   Voltmeter connected across the load, ammeter in series with the source.

Solution

Configuration A

$$I_s = I_L + I_v$$

$$I_L = \frac{V_t}{R_a + R_L} \quad \text{(through ammeter and load)}$$

$$I_v = \frac{V_t}{R_v} \quad \text{(through voltmeter).}$$

Source terminal relation:

$$V_t = V_s - R_s \cdot I_s = \frac{V_s}{1 + R_s\left(\frac{1}{R_a+R_L} + \frac{1}{R_v}\right)}$$

Measured power (what you would calculate from the meter readings, i.e. voltmeter × am-

meter):

$$P_{\text{meas},A} = V_t I_L = \frac{V_t^2}{R_a + R_L} = \frac{V_s^2}{(R_a + R_L)(1 + R_s\left(\frac{1}{R_a+R_L} + \frac{1}{R_v}\right)^2}$$

True power dissipated in load resistor:

$$P_{\text{true}} = I_L^2 R_L = \frac{V_t^2\, R_L}{(R_a + R_L)^2}$$

Configuration B

$$R_p = \frac{R_L R_v}{R_L + R_v} \quad \text{(parallel combination of load and voltmeter)}$$

$$I_s = \frac{V_s}{R_s + R_a + R_p}$$

$$V_L = I_s\, R_p \quad \text{(voltage across load and voltmeter)}$$

Measured power:

$$P_{\text{meas},B} = V_L I_s = I_s^2 R_p = \frac{V_s^2 R_p}{(R_s + R_a + R_p)^2}$$

True power dissipated in load resistor:

$$P_{\text{true}} = I_L^2 R_L$$

$$I_L = \frac{V_L}{R_L} = \frac{I_s R_p}{R_L}$$

$$\Rightarrow P_{\text{true}} = \frac{I_s^2 R_p^2}{R_L}$$

R_L	P_{true}	$P_{\text{meas},A}$	$P_{\text{meas},B}$
$1\,\Omega$	$32.62\,\text{W}$	$35.88\,\text{W}$	$32.65\,\text{W}$
$100\,\Omega$	$1.406\,\text{W}$	$1.407\,\text{W}$	$1.546\,\text{W}$